

Double FW Example

Alejandro De La Torre

October 2025

Due: 11:59pm Central, Friday 10/03/2025. Submit a PDF to Canvas.

1 DOUBLE FW

Using Frisch–Waugh Twice: 2SLS with controls

Setup. Structural equation with one endogenous regressor x_{1i} , exogenous controls $x_{oi} \in \mathbb{R}^q$, and instrument z_i :

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta'_o x_{oi} + u_i, \quad E[u_i \mid z_i, x_{oi}] = 0, \quad (z_i, x_{1i} \mid x_{oi}) \neq 0.$$

First stage (population projection of x_{1i} on instrument and controls):

$$x_{1i} = \pi_0 + \pi_1 z_i + \pi'_o x_{oi} + \eta_i, \quad E[\eta_i \mid z_i, x_{oi}] = 0. \quad (1.1)$$

Let M_{x_o} denote the residual-maker w.r.t. the controls: for any variable w_i ,

$$\tilde{w}_i := (M_{x_o} w)_i \quad (\text{“}w \text{ residualized on } x_o \text{”}).$$

FWL #1 (partial out controls in the first stage). Apply Frisch–Waugh to (1.1) by residualizing both sides on x_o :

$$\tilde{x}_{1i} = \pi_1 \tilde{z}_i + \tilde{\eta}_i, \quad \tilde{z}_i := M_{x_o} z_i, \quad \tilde{x}_{1i} := M_{x_o} x_{1i}. \quad (1.2)$$

The OLS slope from (1.2) is

$$\hat{\pi}_1 = \frac{\sum_i \tilde{z}_i \tilde{x}_{1i}}{\sum_i \tilde{z}_i^2}, \quad \hat{\tilde{x}}_{1i} = \hat{\pi}_1 \tilde{z}_i.$$

The fitted value for x_{1i} from the full first stage is

$$\hat{x}_{1i} = \hat{\tilde{x}}_{1i} + \widehat{\text{proj}}(x_{1i} \mid x_{oi}) = \hat{\pi}_1 \tilde{z}_i + \hat{\pi}'_o x_{oi} + \hat{\pi}_0.$$

Substitution and the second-stage error. Substitute the population first-stage decomposition into the structural equation:

$$\begin{aligned} y_i &= \beta_0 + \beta_1(\pi_0 + \pi_1 z_i + \pi'_o x_{oi} + \eta_i) + \beta'_o x_{oi} + u_i \\ &= (\beta_0 + \beta_1 \pi_0) + \beta_1 \pi_1 z_i + x'_{oi}(\beta_o + \beta_1 \pi_o) + \underbrace{(u_i + \beta_1 \eta_i)}_{=: e_i}. \end{aligned}$$

Hence in the “plug-in” second-stage regression of y_i on $(\hat{x}_{1i}, x_{oi})$, the composite error is

$$e_i = u_i + \beta_1 \eta_i.$$

Under instrument exogeneity and the first-stage orthogonality, $E[z_i e_i | x_{oi}] = 0$ and $E[x_{oi} e_i] = 0$.

FWL #2 (partial out controls in the second stage). By Frisch–Waugh, the OLS coefficient on $\hat{x}_{1i}$ in the regression of y_i on $(\hat{x}_{1i}, x_{oi})$ equals the slope from the regression of the y -residual on the $\hat{x}_1$ -residual:

$$\tilde{y}_i := M_{x_o} y_i, \quad \tilde{\hat{x}}_{1i} := M_{x_o} \hat{x}_{1i}, \quad (1.3)$$

$$\hat{\beta}_1^{2SLS} = \frac{\sum_i \tilde{\hat{x}}_{1i} \tilde{y}_i}{\sum_i \tilde{\hat{x}}_{1i}^2}. \quad (1.4)$$

Using $\hat{x}_{1i} = \hat{\pi}_1 \tilde{z}_i + \widehat{\text{proj}}(x_{1i} | x_{oi})$ and $M_{x_o} x_{oi} = 0$, we have

$$\tilde{\hat{x}}_{1i} = \hat{\pi}_1 \tilde{z}_i.$$

Plug this into (1.4):

$$\hat{\beta}_1^{2SLS} = \frac{\sum_i (\hat{\pi}_1 \tilde{z}_i) \tilde{y}_i}{\sum_i (\hat{\pi}_1 \tilde{z}_i)^2} = \frac{\sum_i \tilde{z}_i \tilde{y}_i}{\hat{\pi}_1 \sum_i \tilde{z}_i^2} = \frac{\sum_i \tilde{z}_i \tilde{y}_i}{\sum_i \tilde{z}_i \tilde{\hat{x}}_{1i}}.$$

Scalar IV form after double partialling. Defining all variables residualized on x_o (tildes), the 2SLS slope can be written as the just-identified IV estimator:

$$\boxed{\hat{\beta}_1^{2SLS} = \frac{\sum_i \tilde{z}_i \tilde{y}_i}{\sum_i \tilde{z}_i \tilde{\hat{x}}_{1i}} = \frac{n(\tilde{z}, \tilde{y})}{n(\tilde{z}, \tilde{\hat{x}}_1)}}$$

and the composite second-stage error is

$$\boxed{e_i = u_i + \beta_1 \eta_i}.$$

Matrix (optional, compact) form. Let $X = [x_1 \ x_o]$, $Z = [z \ x_o]$, $P_Z = Z(Z'Z)^{-1}Z'$. Then the 2SLS estimator is

$$\hat{\beta}_{2SLS} = (X'P_Z X)^{-1} X'P_Z y,$$

and the scalar formula above is the x_1 -coefficient after residualizing both z and (y, x_1) on x_o (FWL twice).

Assumptions used. (i) Relevance: $(\tilde{z}_i) > 0$ and $(\tilde{z}_i, \tilde{x}_{1i}) \neq 0$; (ii) Exogeneity: $E[z_i u_i | x_{oi}] = 0$; (iii) Correct linear projections defining η_i and the FWL residualizations. Under these, $\hat{\beta}_1^{2SLS} \rightarrow_p \beta_1$.